TP : Breaking the Classics: A Cryptography Escape Room

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Mission

At 09:00, the CC@UM6P security team recovers a mysterious encrypted archive from an old communications system. The archive is protected by a chain of classical ciphers, and each message you break reveals the clue needed to unlock the next stage. Your mission: follow the trail, write the right tools, crack every cipher, and recover the final flag before time runs out.

1. Start with Stage 1. Do not start the next challenge file until you have the current access word.
2. For every coding stage, keep the code you used. Your final submission includes the code.

Learning Goals

By the end of this lab, students should be able to:

- gain a practical introduction to cryptography and cryptanalysis;
- encrypt and decrypt messages using well-known classical ciphers such as Caesar, substitution, and Vigenère;
- understand the role of secret keys and key spaces in cryptographic security;
- understand brute-force attacks and identify when they are practical or impractical;
- write simple programs to automate encryption, decryption, and basic cryptanalysis;
- explain why classical ciphers are considered insecure by modern cryptographic standards.

Warm-Up – Before You Enter the Vault

Before the escape room begins, let us make sure your cryptography tools are ready.

This short warm-up does not unlock a room. It is here to refresh Caesar encryption, decryption, and the basic idea of brute force before the mission starts.

The first terminal is still running. On the screen:

1. **Encrypt.**

   Encrypt the following message using a Caesar cipher with K = 4 , Keep spaces unchanged.

   ```
   CRYPTO IS FUN UNTIL THE PROFESSOR SAYS QUIZ
   ```

2. **Decrypt.**

   The following message was encrypted using a Caesar cipher with K = 8

   ```
   GWCL JMBBMZ VWB JM CAQVO BWWTA BW AWTDM BPQA
   ```

   Recover the original plaintext.

3. **The key is gone.**

   A second message appears:

   ```
   ESP VPJ TD FYVYZHY ECJ MCFEP QZCNP
   ```

   This time, you know that the cipher is Caesar...

   but the key has disappeared.

   Instead of testing keys manually, **write a small program** that can help you decrypt the message.

Hint (If needed)

Think like an attacker:
How many possible keys does Caesar actually have?
If the answer is small enough, perhaps you do not need to "discover" the key at all.
You can simply try them all.

- tries every possible Caesar key;
- decrypts the ciphertext using each key;
- displays all candidate plaintexts;

Room 1: The Letter Wall

The warm-up is over. Your first real challenge is waiting inside the archive. A new encrypted message appears on the terminal, together with a short note:

"Shifting the alphabet was too predictable. So I replaced every letter with a different one."

You are now dealing with a **substitution cipher**.

The same plaintext letter is always replaced by the same ciphertext letter.

1. **Learn the mechanism.**

   For the first two tasks, use the following substitution alphabet:

Plain	A	B	C	D	E	F	G	H	I	J	K	L	M
Cipher	Q	M	J	Z	T	G	F	K	P	W	L	S	B

Plain	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	O	X	N	C	R	Y	E	V	H	I	A	D	U

   Encrypt:

   ```
   KEEP THE SECRET SAFE
   ```

2. Using the same substitution alphabet, decrypt:

   ```
   ERVYE DXVR SXFPJ
   ```

3. **Now the substitution alphabet disappears.**

   Could you simply try every possible key? A substitution key may be any permutation of the 26-letter alphabet: 26!

   Brute force is therefore no longer an attractive strategy.

   But the cipher has left something behind: **the structure of the English language.**

   Typical English letter frequencies include:

Letter	E	T	A	O	I	N	S	H	R	D	L	C
Frequency (%)	12.7	9.1	8.2	7.5	7.0	6.7	6.3	6.1	6.0	4.3	4.0	2.8

   Question: How could an attacker use this information to recover part of an unknown substitution key?

   Now turn this idea into a practical attack.

   Write a small program that analyzes a ciphertext by:

   - ignoring spaces and punctuation;
   - counting the occurrences of each letter;
   - calculating the percentage of each letter;
   - displaying the letters from most frequent to least frequent.

   Then use your program to help decrypt the intercepted message below.

   Remember: frequency analysis gives you **clues**, not necessarily the complete key. Combine your results with repeated words, short words, letter patterns, and context.

4. **Break the intercepted message.**

   The following ciphertext was recovered from the archive. The substitution alphabet is unknown.

```
EKT YTJRTE BTYYQFT PY KPZZTO PO EKPY NQRQFRQNK.
STEETR GRTCVTOJD ZXTY OXE RTHTQS EKT IKXST LTD,
MVE PE FPHTY Q FXXZ YEQRE.
SXXL GXR JXBBXO IXRZY, RTNTQETZ NQEETROY,
QOZ YKXRE IXRZY.
IKTO DXV RTJXHTR EKT BTYYQFT,
EKT HPFTOTRT LTD KQY GXVR STEETRY.
```

Use your frequency-analysis program to recover the plaintext and the clue for the next room. Frequency analysis usually gives you a *hypothesis*, not the exact answer.

Hint (If needed)

For this message, you may assume the frequent plaintext letters occur roughly in this order:

$$E > T > R > S/O > A > H > N > D > I > G > L$$

The symbol "/" means that the two letters occur with approximately the same frequency. Compare this with your ciphertext frequencies and build a partial mapping.

Room 2 – Break the Vigenère Cipher

You intercepted the following ciphertext:

```
VVHWGQUIVYHIRSUPGTWXJSVIPCWIUPHGCIVIVVHPGHWITSDTRSDVUSYITMZL
GFHIXSUCVSDQOSPFGFVLQIOHEVHGMSDGJQRPWAQWGSNXJSPSUHIVGEXIPHVC
OPRPCBGYUSLXVCUIECYITHKIMSBAJSQXJSPIUGDKGPHGQAHWEZHETSQXGFWL
GOFGGGVAQFGIEZLTUS
```

You know that:

- the cipher is **Vigenère**;
- the plaintext is written in English;
- an important clue about the key was recovered in the previous room.
- Tthe most frequent plaintext letter in each group is likely to be:

Your mission is to:

1. recover the **key**;
2. decrypt the message;
3. find the hidden **access word**.

You may reuse the frequency-analysis program developed in the previous exercise.

UP TO THE CHALLENGE?

Stop here and try to solve it by yourself.

Only continue below if you need help.
Reveal the hints one by one.

Hint 1 – Follow the repeating key

A 4-letter key repeats continuously:

```
K1 K2 K3 K4 K1 K2 K3 K4 K1 K2 K3 K4 ...
```

Using Python indexing, this means:

Key letter	Ciphertext positions
K1	0, 4, 8, 12, 16, ...
K2	1, 5, 9, 13, 17, ...
K3	2, 6, 10, 14, 18, ...
K4	3, 7, 11, 15, 19, ...

Therefore, split the ciphertext into four groups:

```
group1 = ciphertext[0::4]
group2 = ciphertext[1::4]
group3 = ciphertext[2::4]
group4 = ciphertext[3::4]
```

Hint 2 – Something familiar appears

All letters inside one group were encrypted using the **same key letter**.

Therefore, all letters in that group were shifted by the **same amount**.

Each group can therefore be treated like a cipher you already know:

a Caesar cipher.

Hint 3 – Reuse your previous tool

Reuse the frequency-analysis program from the previous exercise.

Run it separately on each of the four groups.

For this challenge, you may assume that the most frequent plaintext letter in each group is likely to be:

E

Hint 4 – Recover the key

For each group:

Most frequent ciphertext letter

↓

Assume that it corresponds to plaintext **E**

↓

Determine the Caesar shift

↓

Convert that shift into a key letter

For example, if the most frequent ciphertext letter is `G`:

$$E \rightarrow G$$

the shift is 2, therefore the corresponding key letter is:

$$\boxed{C}$$

(Check Appendix)

Repeat the process for all four groups, combine the four recovered letters, and use the resulting key to decrypt the complete ciphertext.

Ciphertext Appendix

Vigenère Table

Appendix – Vigenère Quick Reference

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y